

9(a)(i) Resistance = $\frac{\rho L}{A}$ [1]

$$= \frac{\rho L}{\left(\frac{\pi d^2}{4} \right)}$$

Cross sectional area of wire, $A = \frac{\pi d^2}{4}$

$$= \frac{1.50 \times 10^{-6} \times 6.0 \times 10^{-2}}{\left(\frac{\pi (0.30 \times 10^{-3})^2}{4} \right)}$$

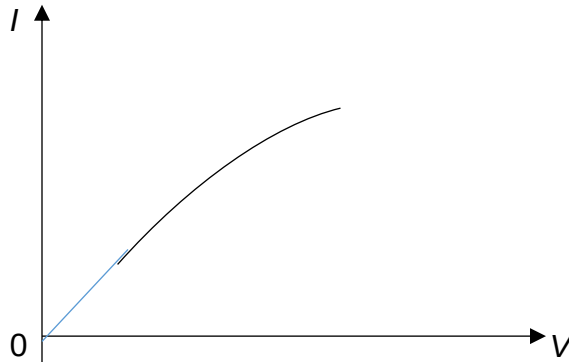
$$= 1.273 \quad [1]$$

$$= 1.3 \, \Omega$$

- (ii) 1. e.m.f. is the amount of other forms of energy converted to electrical energy per unit charge delivered by a source of e.m.f. [1]

p.d. is the amount of electrical energy converted to other forms of energy per unit charge flowing through a device. [1]

2.



[1; no need to show negative quadrant]

3. Read off the corresponding value of I .

$$\text{Resistance} = \frac{V}{I} \quad [1; \text{must have both statements}]$$

4. Fraction of power delivered = $\frac{\text{Power dissipated through X}}{\text{Total power dissipated in circuit}}$

$$= \frac{I^2 R_X}{I^2 (R_X + R_{\text{variable}})} \quad (R_X \text{ and } R_{\text{variable}} \text{ are in series})$$

[1]

$$= \frac{R_X}{(R_X + R_{\text{variable}})}$$

$$= \frac{1.3}{(1.3 + 0.50)}$$

$$= 0.72 \quad [1]$$

(b)(i)

- As the resistance of the rheostat is increased, the total resistance across the voltmeter is increased. [1]
- Using the potential divider principle, the p.d. across the voltmeter will take up a bigger fraction of the e.m.f. [1]
- So the voltmeter reading will increase. [1]

(ii) S is to prevent short circuit of the cell [1] when the rheostat is set to $0 \, \Omega$. [1]

(iii)1. If voltmeter reads 1.2 V, then p.d. across S = $3.0 - 1.2$
 $= 1.8 \, \text{V}$ [1]

$$\begin{aligned} \therefore \text{current delivered by cell} &= \text{current through S} \\ &= 1.8 / 0.60 \\ &= 3.0 \, \text{A} \quad [1] \end{aligned}$$

2. Current through rheostat = $1.2 / 10$
 $= 0.12 \, \text{A}$ [1]

$$\begin{aligned} \text{Current through bulb} &= \text{Main current} - \text{current through rheostat} \quad [1] \\ &= 3.0 - 0.12 \\ &= 2.9 \, \text{A} \quad [1] \end{aligned}$$

3.

- When the rheostat is set to $0 \, \Omega$, a current will still flow through S. [1]
- So power will be wasted [1], that's why the suggestion is not practical.